

IOS vs GenOS: Positioning, Differences, and Convergence Path

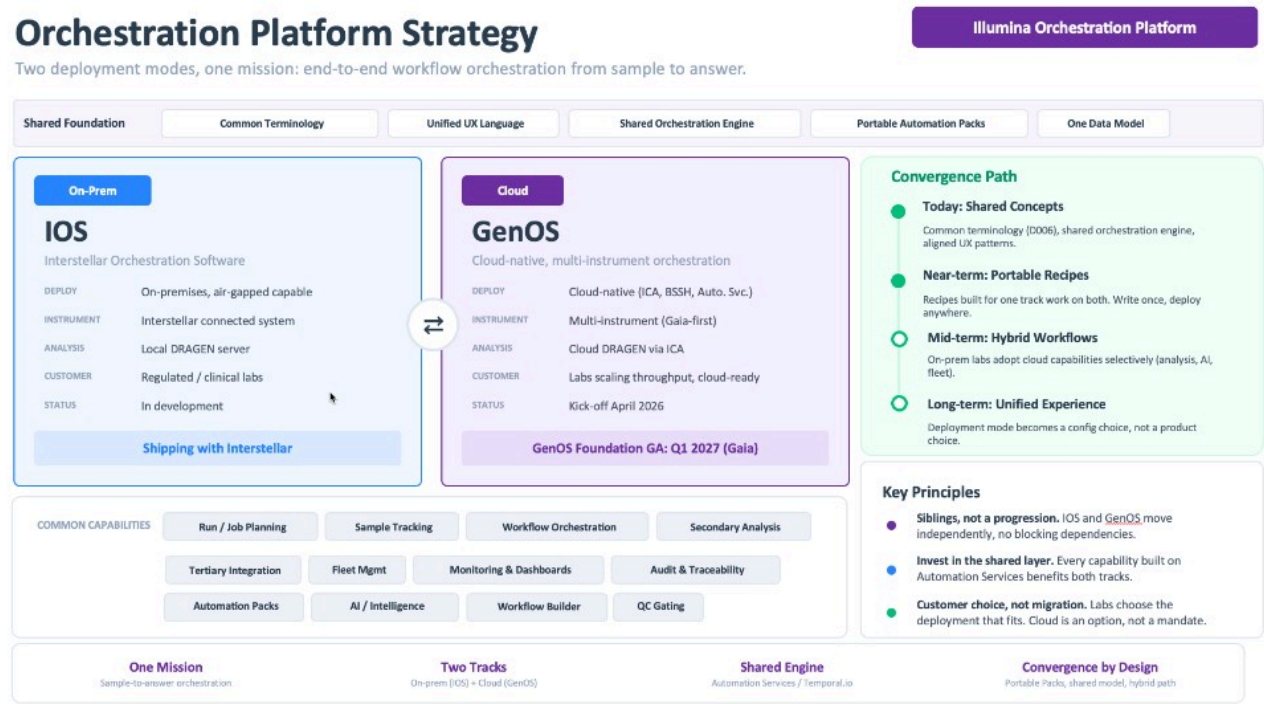
Last updated: 2026-04-06 **Audience:** All team members working on IOS, GenOS, or Gaia
Purpose: Single reference to understand how IOS and GenOS relate, where they differ, and how they converge over time.

The One-Liner

IOS and GenOS are **sibling products** in Illumina's orchestration portfolio. They share a conceptual model and core terminology but have independent architectures, each optimized for a different deployment model and customer profile.

- **IOS** = On-prem orchestration for the Interstellar connected system
- **GenOS** = Cloud-native orchestration for multi-instrument labs (Gaia-first, extensible)

They are not a progression. One does not replace or evolve into the other.



Why Two Products?

GenOS was originally framed as evolving from IOS ("GCO -> IOS -> GenOS"). As the GenOS Foundation scope solidified for the Gaia Q1 2027 co-launch, it became clear that building on IOS's on-prem architecture would slow execution and constrain both products.

The decision to make them siblings unlocks:

- **Parallel execution.** IOS and GenOS teams move independently with no blocking dependencies.
- **Cloud service reuse.** GenOS builds on mature production infrastructure (BioInsight Platform Core, DRAGEN cloud, Automation Services, BaseSpace, ICM, Connected Insights) instead of recreating these for on-prem.
- **Fewer architectural constraints.** IOS's air-gapped design imposes connectivity, update delivery, and compute scaling constraints that don't apply to GenOS. Forcing both through one architecture would compromise both.
- **Broader reach.** IOS targets Interstellar; GenOS targets the much larger installed base of all Illumina sequencers.

Comparison Table

Dimension	IOS	GenOS (Foundation - Q1 2027)
One-liner	Orchestration layer for Interstellar	Cloud-native dry lab orchestration for multi-instrument labs
Deployment	On-prem, air-gapped capable	Cloud-native (runs on BioInsight Platform Core)
Instrument scope	Interstellar connected system	Multi-instrument (Gaia at launch, extensible to NovaSeq X, NextSeq, MiSeq)
Analysis engine	Local DRAGEN server	Cloud DRAGEN via BioInsight Platform Core
Connectivity	Cloud-connected optional (for tertiary analysis)	Cloud-required
Target customer	Regulated/clinical labs needing on-prem or air-gapped operation	Labs comfortable with cloud, scaling throughput
User experience	Task-oriented operator UI + admin console, locally hosted, no internet required	Cloud UX with self-service onboarding, cross-instrument visibility, and workflow builder
Integrations	LIS/LIMS/EHR (through 3rd party vendor), file-based patterns, local	TBD

Dimension	IOS	GenOS (Foundation - Q1 2027)
	APIs	
Tertiary handoff	ICM (local or cloud)	TBD
Fleet management	Future roadmap item	Multi-instrument fleet visibility from day one
AI capabilities	Future roadmap item	Built in from v1: Predictive QC, Smart Batching, Anomaly Detection
Workflow model	Pre-configured validated menus (fixed step categories)	User-configurable Steps with chainable Actions from the Action Library
Update delivery	On-prem software updates (strategy TBD)	Continuous cloud releases
Timeline	Wave 1 shipping 2027	GenOS Foundation GA Q1 2027 (Gaia co-launch)

What They Share

IOS and GenOS are independent architectures, but they share enough to feel like part of one portfolio:

Shared Element	Details
Terminology	Workflow, Step, Recipe, Action, Execution, Trigger. Same terms, same meanings, across both products, UI, docs, and customer conversations.
Orchestration engine	Automation Services (Temporal.io runtime) is the core execution engine for both. Actions built for one product can run on the other.
Conceptual model	Both follow the same layered hierarchy: Workflow -> Steps -> Actions (with Recipe as the container for Action chains).
Data model concepts	Specimen, Sample, Library, Run, Flow Cell, Instrument, Case. Same entities, same definitions.
Design language direction	Both products are working toward a unified UX language over time.

What IOS Is NOT

- **Not GenOS.** IOS serves on-prem/Interstellar; GenOS serves cloud/multi-instrument.
- **Not a full LIMS.** IOS has LIMS-like surfaces (sample tracking, statuses) but is not positioned as a LIMS replacement.
- **Not going away.** Air-gapped labs need on-prem. IOS serves that need.

What GenOS Is NOT

- **Not IOS.** GenOS is not built on IOS and does not require IOS.
- **Not on-prem capable.** GenOS Foundation is cloud-only. On-prem is an IOS capability.
- **Not a replacement for IOS.** Labs needing air-gapped or fully disconnected operation use IOS.

Customer Decision Framework

If the lab...	They use...
Runs Interstellar and needs on-prem or air-gapped operation	IOS
Is in a regulated environment that cannot use cloud	IOS
Runs Gaia, NovaSeq X, NextSeq, MiSeq, or other Illumina sequencers	GenOS
Is comfortable with cloud and wants multi-instrument visibility	GenOS
Wants self-service onboarding without Professional Services	GenOS

Convergence Path

IOS and GenOS are independent today, but they converge over time through shared building blocks, not by merging into one product.

Now: Shared Vocabulary and Engine

Both products use the same terminology and the same orchestration engine (Automation Services / Temporal.io). This means teams speak the same language, docs are consistent, and the core execution model is identical.

Near-term: Portable Actions

Actions built for one product work on the other. "Write once, deploy to cloud or on-prem." This is the primary convergence mechanism: the same automation logic runs in both deployment models.

Mid-term: Shared UX Patterns

As both products mature, the UX layer will converge toward shared design patterns and components. A lab operator moving between IOS and GenOS should find a familiar experience.

Long-term: Deployment as Configuration

The aspiration is a converged experience where deployment mode (cloud vs. on-prem) becomes a configuration choice, not a product choice. This is directional, not committed.

Guiding Principles

- **Siblings, not a progression.** IOS and GenOS move independently. No blocking dependencies.
 - **Customer choice, not migration.** Labs choose the deployment that fits. The path to cloud is an option, not a mandate.
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Common Misconceptions

Misconception	Reality
"GenOS replaces IOS"	No. They are siblings serving different deployment models. Both continue.
"IOS is the old version, GenOS is the new version"	No. IOS is purpose-built for on-prem/Interstellar. GenOS is purpose-built for cloud/multi-instrument. Neither is a version of the other.
"We should build on IOS first, then port to GenOS"	Building on IOS's on-prem architecture would slow GenOS and constrain both. Each builds on the architecture best suited to its deployment model.
"GenOS can run on-prem"	Not in v1. On-prem is an IOS capability. GenOS Foundation is cloud-only.
"IOS is only for Interstellar forever"	IOS is Interstellar-first, but its architecture is modular. Expansion decisions will come with Wave 2+.

Misconception	Reality
"The teams should merge since it's the same thing"	Same conceptual model, independent architectures. The teams share the orchestration engine (Automation Services) and terminology, but product development is parallel.